Detecting anomalous activity of a ship's engine

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1 Introduction

Many sectors are employing advancements in AI and leveraging historical data to reduce maintenance costs, predict equipment breakdown, reduce emissions and prevent failures. Logistics sector that uses numerous fleets are no different especially when they own large transportation vehicles such as cargo planes and ships. Having a reliable resources is crucial for any business to maintain good customer retention rate and reducing churn rate.

2 Problem statement

Inadequate maintenance of ship engines can lead to disastrous consequences for businesses in the supply chain industry such as loss in turnover and increased churn rate. Predicting abnormal behaviour of ship engine in early stages can help businesses to implement data-driven strategies. Statistical methods and ML algorithms can predict unusual ship engine performance by analysing parameters such as Engine's revolutions per minute, coolant pressure, lubricant oil temperature etc.

3 Dataset

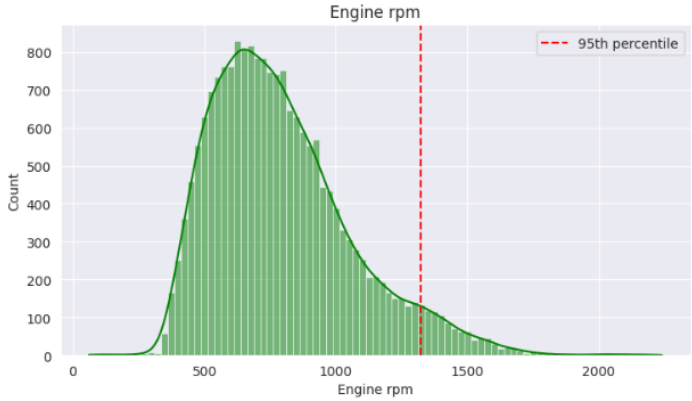
The ship engine's dataset was given in the course material and extracted from github [1]. The dataset has 19535 rows and 6 columns (features) namely

- Engine's rpm
- Lubricant oil pressure
- Fuel pressure
- Coolant pressure
- Lubricant oil temperature
- Coolant temperature

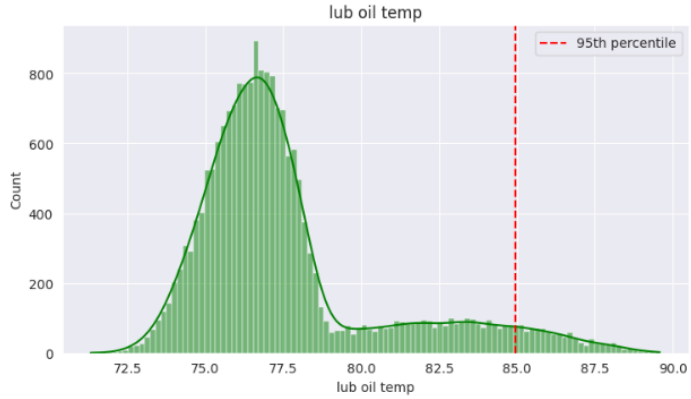
There are no missing or duplicated data in the dataset and the data types of the features are in decimals (float64) except for Engine rpm which is a integer type (int64). Descriptive statistics revealed that 95% of the data have values below 1324 for the Engine's rpm, 5.06 for lubricant oil pressure, 12.21 for fuel pressure, 4.44 for coolant pressure, 84.94 for lubricant oil temperature, and 88.61 for coolant temperature. Any values above can be potentially considered as anomalies.

3.1 Data Distribution

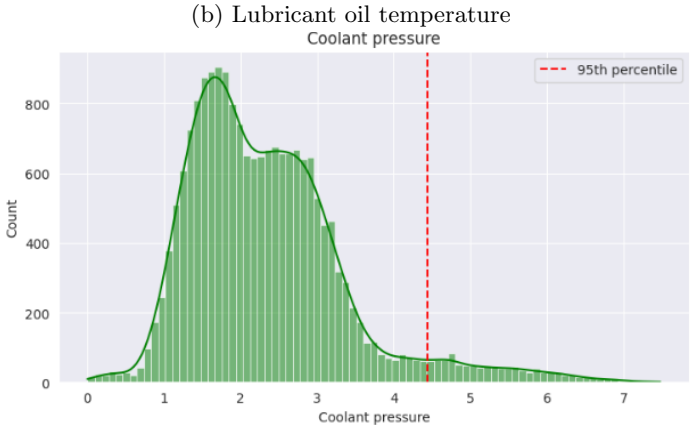
From the histogram plot of the features in figure 1, we can clearly see that the data is not normally distributed but rightly skewed. This means there are anomalies beyond the upper bound which is confirmed by the box plots in figure 2.



(a) Engine rpm

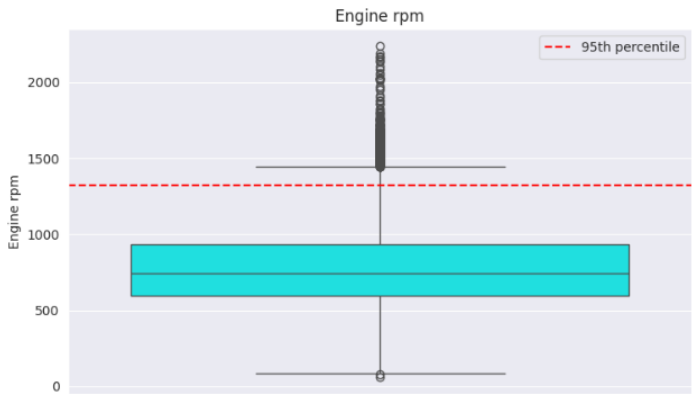


(b) Lubricant oil temperature

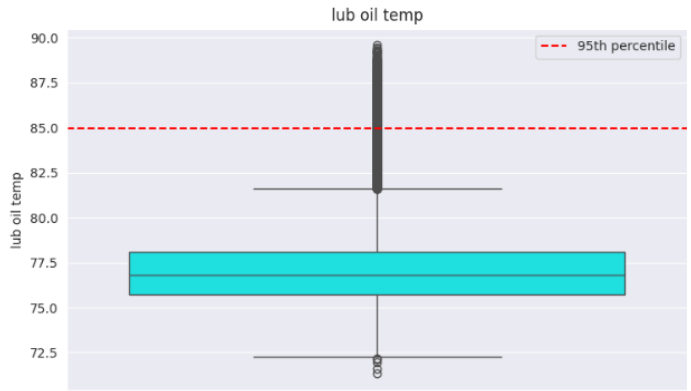


(c) Coolant pressure

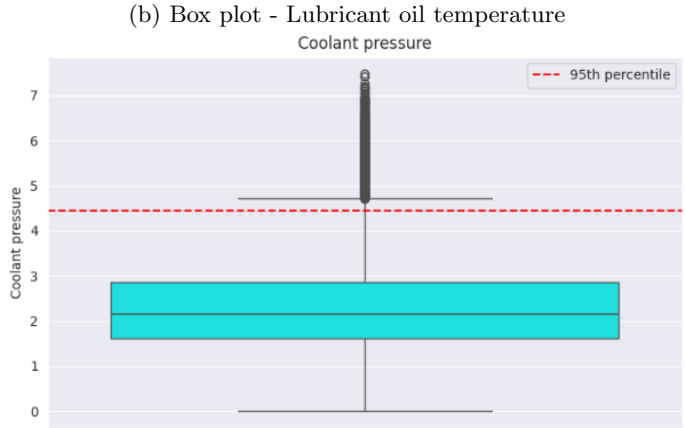
Figure 1: Data distribution using histogram



(a) Boxplot - Engine rpm



(b) Box plot - Lubricant oil temperature



(c) Boxplot - Coolant pressure

Figure 2: Data distribution using histogram

4 Methods

4.1 Anomaly detection via statistical methods

To identify outliers for all features in the dataset, I employed the inter-quartile range (IQR) method. 25th and 75th percentiles are calculated to formulate upper and lower bounds for each feature. A binary column is created where 0 is assigned to be a normal value and 1 is a outlier. These binary columns are then appended to the dataset and a total-outlier column is created which sums the binary columns. Then the dataset is filtered on total-outliers column that is greater than or equal to 1, 2 and 3 respectively.

4.2 Anomaly detection via Machine Learning models

For One Class Support Vector Machine (OCSVM), the dataset is scaled and feed into the model with initial parameters of gamma = 0.5 and nu = 0.05 while kernel function is set to 'rbf'. To visualise the model, Principal Component Analysis (PCA) was used to reduce the dimensions of the dataset from 6 to 2 features as shown in figure 3a. Several models were created for varying values of gamma and nu and the results were recorded. Isolated Forest (IF) model was fed with un-scaled dataset with a contamination value = 0.05 and number of estimators at 100. Similar to OCSVM, parameters were fine tuned and the results were visualised through PCA as shown in figure 3b.

5 Results

5.1 IQR results

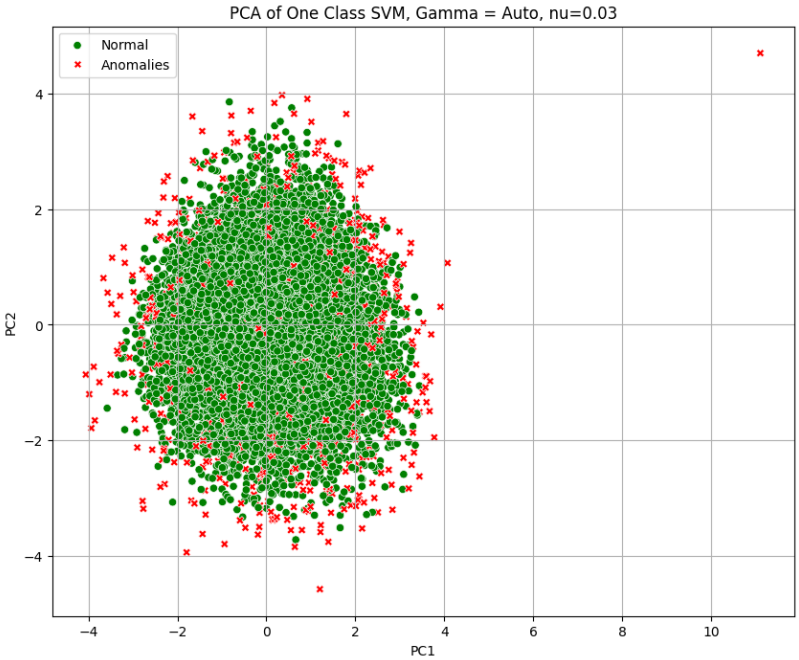
IQR method predicted that there are 4634 anomalies (23.72%) when binary columns are 1 or greater, 422 anomalies (2.16%) for 2 or greater and 11 anomalies (0.056%) for 3 or greater columns. Since 23.72% and 0.056% are outside the expected anomalies range, 2.16% anomalies when 2 or greater binary columns is considered to be a good estimator for this dataset.

5.2 ML model results

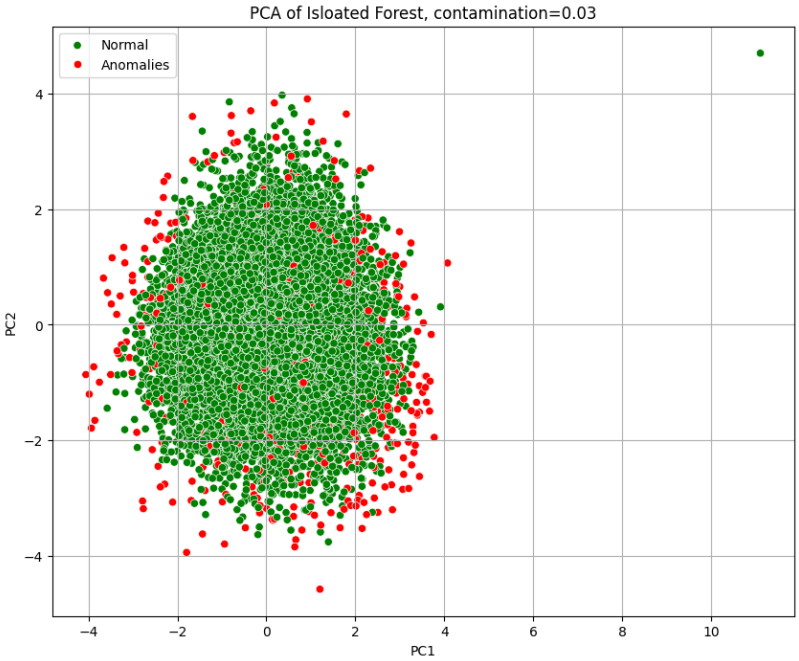
OCSVM model predicted 1043 anomalies resulting in 5.34% outside the expected range. After a few trails of fine tuning gamma and nu to 'auto' and 0.03, the model predicted 2.99 % (585 anomalies). IF model with N-estimators 100 and contamination rate 0.05 predicted 5.00% (977 anomalies). Higher contamination rates such as 0.07 and 0.1 predicted 7% and 10% outliers which are outside expected range. Contamination rate of 0.03 predicted 3.0 % (587 anomalies) at N-estimators 100. Varying the N-estimators had no impact on the outliers. Both OCSVM and IF model predicted approximately the same number of outliers (3%).

5.3 Common anomalies

When comparing IQR method with OCSVM, 171 rows are common. 358 rows are common to both OCSVM and IF method and lastly 221 rows are common to IF and IQR method. Approximately both OCSVM and IF models predicted 3% of anomalies (587 rows) in the dataset tolerating within expected range of 1-5%. Using intersection method, 154 anomalies are common to all three methods.



(a) PCA of OCSVM model



(b) PCA of IF model

6 Conclusion

For the given dataset, OCSVM model with gamma as auto, nu = 0.03 predicted anomalies within the expected range. Likewise, IF model with contamination rate = 0.03 at n-estimators 100, predicted anomalies within the given range. If both these models predicted 587 anomalies (3%) and out of these 358 anomalies are mutual accounting for 61 % between the models. Similarity between OCSVM and IF models are visualised in PCA plots shown in figures 3a and 3b.

However, the foundation of every data science project is Business Understanding. In this project, as I am neither an expert in marine engineering nor possess business understanding of ship's engine, I can only follow where the data is taking me. Ideally, I would like to consult my findings with ship engineers who are the domain experts. These domain experts could argue and classify some anomalies as normal data points and vice versa.

References

[1] `https://raw.githubusercontent.com/fourthrevlxd/cam_dsb/main/engine.csv`